LINEAR ALGEBRA

MATH 105

TOPIC: MATRIX ALGEBRA

Lecture 1

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OUTLINE OF LECTURE

- Introduction
- Matrix Definition
- Types of Matrices
- Transpose and Symmetric Matrices
- Matrix Operations
  - ✓ Equality of matrices
  - ✓ Addition and subtraction of matrices

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Introduction

- Linear Algebra is a branch of mathematics that involves linear relationships of variables and numbers
- The term consists of two words
  - ✓ Linear: the relationship gives a straight line plot on the xy-plane
  - ✓ Algebra: deals with variables (symbols) and the rules for manipulating those symbols
- Full appreciation of the meaning is expected at the end of the course

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Introduction

- Linear Algebra has a lot of applications in Computer Science and related areas
  - ✓ Computer Graphics: Matrix transformations, rotations and scale operations in 3D animation and image processing
  - ✓ Artificial Intelligence: matrices and vectors for neural networks
  - ✓ Machine Learning: Matrices and vectors are applied in the development of algorithms

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Introduction

- ✓ Data Science: Linear algebraic techniques for large dataset manipulation
- ✓ Computer Vision: Matrix transformations and rotations for feature extraction and pattern recognition
- ✓ Cryptography: Application of matrices for encryption and decryption algorithms
- ✓ Signal Processing: areas of signal analysis, compression and filtering use linear algebra applications
- ✓ Etc

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Matrix Definition

- Matrix is the rectangular array of numbers, variables or functions
- Each of the numbers (variables) in the matrix is called an ***entry (element)***
- The size of a matrix is determined by the number of rows and number of columns
- A matrix with $\boldsymbol{m}$ rows and $\boldsymbol{n}$ columns is denoted by $\boldsymbol{m} \times \boldsymbol{n}$
- Matrices are denoted by capital (upper case) letters

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Matrix Definition

- For a matrix ***A***, the entry in ***mth row*** and ***nth column*** is denoted by $\boldsymbol{a_{mn}}$
- It is generally represented as

$$
\begin{array}{cc}
 & \text{Column } j \\
\text{Row } i & \begin{bmatrix} a_{11} & \cdots & a_{1j} & \cdots & a_{1n} \\ \vdots & & \vdots & & \vdots \\ a_{i1} & \cdots & a_{ij} & \cdots & a_{in} \\ \vdots & & \vdots & & \vdots \\ a_{m1} & \cdots & a_{mj} & \cdots & a_{mn} \end{bmatrix} = A \\
 & \begin{array}{ccc} \uparrow & \uparrow & \uparrow \\ \mathbf{a}_1 & \mathbf{a}_j & \mathbf{a}_n \end{array}
\end{array}
$$

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Matrix Definition

Order/Size/Dimension of a Matrix

- The order $\boldsymbol{m} \times \boldsymbol{n}$ of a matrix is the size indicating the number of **m** rows and **n** columns

Illustration

$$\mathbf{A} = [a_{ij}] = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}.$$

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Matrix Definition

- Examples are

$1 \times 1 \qquad [3]$ (scalar 3)

$1 \times 2 \qquad \begin{bmatrix} 3 & 2 \end{bmatrix}$

$2 \times 1 \qquad \begin{bmatrix} 5 \\ -1 \end{bmatrix}$

$2 \times 2 \qquad \begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix}$

$2 \times 3 \qquad \begin{bmatrix} 2 & 0 & 1 \\ -1 & 3 & -2 \end{bmatrix}$

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Types of Matrices

Row Matrix

- Row matrix has only one (single) row denoted as $\mathbf{1} \times \boldsymbol{n}$
- Examples are

$$1 \times 2 \qquad \begin{bmatrix} 3 & 2 \end{bmatrix}$$

$$1 \times 3 \qquad \begin{bmatrix} 5 & 0 & -1 \end{bmatrix}$$

$$1 \times 4 \qquad \begin{bmatrix} 2 & -1 & 2 & 1 \end{bmatrix}$$

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Types of Matrices

Column Matrix

- Column matrix has only one (single) column denoted as $\boldsymbol{m} \times \mathbf{1}$
- Examples are

$$2 \times 1 \qquad \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$3 \times 1 \qquad \begin{bmatrix} 2 \\ 0 \\ 3 \end{bmatrix}$$

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Types of Matrices

Vector

- It is a special type a matrix with either one row or one column
- The entries of a vector are called the components
- Vectors are denoted by small (lowercase) bold letters
- Vectors are classified as
  - ✓ Row vectors
  - ✓ Column vectors

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Types of Matrices

- Examples of row vectors:

  a = (3 5 -2)

  b = $(1 \quad 0 \quad 1)$,

  c = $[1 \quad 6 \quad 0 \quad 6 \quad 8]$

- Examples of column vectors:

$$\mathbf{a} = \begin{pmatrix} 3 \\ 6 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 2 \\ 0 \\ 4 \end{pmatrix}, \quad \mathbf{c} = \begin{pmatrix} 3 \\ 1 \\ 9 \\ 6 \end{pmatrix}$$

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Types of Matrices

Zero Matrix

- Zero matrix has all entries as zeros denoted as $\mathbf{0}_{mn}$
- Examples are

$$1 \times 2 \qquad \begin{bmatrix} 0 & 0 \end{bmatrix}$$

$$2 \times 1 \qquad \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$2 \times 2 \qquad \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

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Types of Matrices

Rectangular Matrix

- It is a matrix with different number of rows and columns denoted as $\boldsymbol{a_{mn}}$ $(m \times n)$
- Examples are

$$2 \times 3 \qquad \begin{bmatrix} 1 & 0 & -1 \\ 3 & 2 & 4 \end{bmatrix}$$

$$3 \times 2 \qquad \begin{bmatrix} 4 & 5 \\ 1 & 3 \\ 0 & 1 \end{bmatrix}$$

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Types of Matrices

Square Matrix

- It is a matrix with equal number of rows and columns denoted as $\boldsymbol{a_{mm}}$ $(m \times m)$ or $\boldsymbol{a_{nn}}$ $(n \times n)$
- Examples are

$$2 \times 2 \qquad \begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix}$$

$$3 \times 3 \qquad \begin{bmatrix} 2 & 0 & 1 \\ -1 & 1 & 3 \\ 3 & 2 & 1 \end{bmatrix}$$

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Types of Matrices

Leading/Main Diagonal

- They are the elements in the top left diagonal to bottom right of a square matrix
- The entries are: $a_{11}, a_{22}, a_{33}, \ldots, a_{mm}$

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1m} \\ a_{21} & a_{22} & \cdots & a_{2m} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nm} \end{bmatrix}$$

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Types of Matrices

Trace of a Matrix

- The **trace** of a square matrix A written as $\boldsymbol{tr(A)}$ is the sum of the leading diagonal elements of the matrix
- Given

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \cdots & a_{nn} \end{bmatrix}$$

$$\boldsymbol{tr(A) = a_{11} + a_{22} + a_{33} + \cdots + a_{nn}}$$

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Types of Matrices

Illustration

Find the trace of the following matrices

$$A = \begin{bmatrix} 1 & 3 \\ 3 & 4 \end{bmatrix}, K = \begin{bmatrix} 2 & 0 \\ 2 & -1 \end{bmatrix}, P = \begin{bmatrix} 4 & 1 & 0 \\ -1 & 1 & 2 \\ 3 & 2 & -3 \end{bmatrix}$$

Solution

$$tr(A) = 1 + 4 = 5$$

$$tr(K) = 2 + (-1) = 2 - 1 = 1$$

$$tr(P) = 4 + 1 + (-3) = 5 - 3 = 2$$

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Types of Matrices

Triangular Matrix

- It is a square matrix where all entries above or below the main diagonal are zeros
- It is classified as
  - ✓ Upper Triangular matrix
  - ✓ Lower Triangular matrix

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Types of Matrices

Upper Triangular matrix

- It has nonzero entries only on and above the main diagonal with all other entries below the main diagonal as zeros.
- Examples

$$A = \begin{pmatrix} 5 & 2 \\ 0 & 3 \end{pmatrix}, B = \begin{bmatrix} 1 & 0 & 3 \\ 0 & 3 & 1 \\ 0 & 0 & 2 \end{bmatrix}, C = \begin{pmatrix} 3 & 1 & 2 & 0 \\ 0 & 7 & 3 & 1 \\ 0 & 0 & 4 & 5 \\ 0 & 0 & 0 & 2 \end{pmatrix}$$

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Types of Matrices

Lower Triangular matrix

- It has nonzero entries only on and below the main diagonal with all other entries above the main diagonal as zeros
- Examples

$$A = \begin{pmatrix} 4 & 0 \\ 3 & 5 \end{pmatrix},\ B = \begin{pmatrix} 6 & 0 & 0 \\ 1 & 7 & 0 \\ 5 & 3 & 4 \end{pmatrix},\ C = \begin{pmatrix} 3 & 0 & 0 & 0 \\ 2 & 7 & 0 & 0 \\ -4 & 2 & 1 & 0 \\ 1 & -6 & 4 & 2 \end{pmatrix}$$

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Types of Matrices

Diagonal matrix

- It is a square matrix with nonzero entries on the main diagonal and all other entries as zeros
- Examples

$$A = \begin{pmatrix} \mathbf{1} & 0 \\ 0 & \mathbf{3} \end{pmatrix}, \quad B = \begin{bmatrix} \mathbf{2} & 0 & 0 \\ 0 & \mathbf{4} & 0 \\ 0 & 0 & \mathbf{1} \end{bmatrix}, \quad C = \begin{pmatrix} \mathbf{3} & 0 & 0 & 0 \\ 0 & \mathbf{7} & 0 & 0 \\ 0 & 0 & \mathbf{1} & 0 \\ 0 & 0 & 0 & \mathbf{2} \end{pmatrix}$$

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Types of Matrices

Identity (Unit) Matrix

- It is denoted as $\boldsymbol{I_{mm}}$ or $\boldsymbol{I_m}$
- It is a square matrix with main diagonal elements of 1 and all other elements 0
- It is a special type of diagonal matrix where the diagonal elements are all equal to 1
- A matrix I is an identity matrix if its product with any matrix A, gives the matrix A

$$\boldsymbol{IA = AI = A}$$

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Types of Matrices

- Examples of Identity matrix

$$I_2 = I_{22} \qquad \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$I_3 = I_{33} \qquad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

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Transpose and Symmetric Matrices

- For a matrix $\boldsymbol{A}$ $(\boldsymbol{a_{ij}})$ with order $(\boldsymbol{m} \times \boldsymbol{n})$
  - ✓ Transpose of $\boldsymbol{A}$ denoted as $\boldsymbol{A^T}(\boldsymbol{a_{ji}})$
  - ✓ It is formed by interchanging the rows and columns of $\boldsymbol{A}$ to produce $(\boldsymbol{n} \times \boldsymbol{m})$ matrix
- For the entries

$$\boldsymbol{A^T} = \left(\boldsymbol{a_{ij}}\right)^{\boldsymbol{T}} = (\boldsymbol{a_{ji}})$$

- The order changes

$$(\boldsymbol{m} \times \boldsymbol{n})^{\boldsymbol{T}} = (\boldsymbol{n} \times \boldsymbol{m})$$

- Note that, transposition converts row vectors to column vectors and vice versa

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Transpose and Symmetric Matrices

Illustration

a.

If $A = \begin{bmatrix} 2 & 6 & 3 \\ 1 & 0 & 4 \end{bmatrix}$ then $A^T = \begin{bmatrix} 2 & 1 \\ 6 & 0 \\ 3 & 4 \end{bmatrix}$, and if $A = [7, 3, 2]$ then $A^T = \begin{bmatrix} 7 \\ 3 \\ 2 \end{bmatrix}$.

b.

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, A^T = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$$

$$B = \begin{bmatrix} 4 & 10 & -7 \\ 5 & -1 & 3 \end{bmatrix}, B^T = \begin{bmatrix} 4 & 5 \\ 10 & -1 \\ -7 & 3 \end{bmatrix}$$

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Transpose and Symmetric Matrices

Example

Find the transpose of the following matrices.

i. $A = \begin{pmatrix} 3 & 2 & 4 \end{pmatrix}$ ii. $B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

iii. $C = \begin{pmatrix} 2 & 6 & 3 \\ 8 & 5 & 9 \end{pmatrix}$ iv. $D = \begin{pmatrix} 4 & -5 \\ 1 & 3 \\ 2 & 7 \end{pmatrix}$

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Transpose and Symmetric Matrices

Solution

i. A = (3 2 4), $A^T = \begin{pmatrix} 3 \\ 2 \\ 4 \end{pmatrix}$

ii. B = $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $B^T = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$

iii. C = $\begin{pmatrix} 2 & 6 & 3 \\ 8 & 5 & 9 \end{pmatrix}$, $C^T = \begin{bmatrix} 2 & 8 \\ 6 & 5 \\ 3 & 9 \end{bmatrix}$

iv. D = $\begin{pmatrix} 4 & -5 \\ 1 & 3 \\ 2 & 7 \end{pmatrix}$, $D^T = \begin{pmatrix} 4 & 1 & 2 \\ -5 & 3 & 7 \end{pmatrix}$

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Transpose and Symmetric Matrices

Properties for transposition

- $(A^T)^T = A$
- $(A + B)^T = A^T + B^T$
- $(kA)^T = kA^T$, for a scalar k
- $(AB)^T = B^T A^T$

Prove the above properties given that

$$A = \begin{pmatrix} 2 & 1 & 4 \\ 0 & 3 & 2 \\ 5 & 1 & 8 \end{pmatrix}, \quad B = \begin{pmatrix} 3 & 1 & 2 \\ 6 & 2 & 0 \\ 3 & 4 & 1 \end{pmatrix} \text{ and } c = \frac{1}{2}$$

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Transpose and Symmetric Matrices

Symmetric Matrix

- It is a square matrix whose transpose is the same as the original matrix
- Thus, a matrix **A** is said to be symmetric if and only if $\boldsymbol{A}^{\boldsymbol{T}} = \boldsymbol{A}$
- For entries

$$\boldsymbol{a}_{\boldsymbol{ij}} = \boldsymbol{a}_{\boldsymbol{ji}}$$

for all $\boldsymbol{i}$ and $\boldsymbol{j}$

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Transpose and Symmetric Matrices

Example

Find the transpose of the following matrices

i. $A = \begin{pmatrix} 2 & 3 \\ 3 & 4 \end{pmatrix}$, ii. $B = \begin{pmatrix} 3 & 1 & 4 \\ 1 & 5 & 2 \\ 4 & 2 & -1 \end{pmatrix}$, iii. $C = \begin{pmatrix} 4 & 5 & 1 & 2 \\ 5 & 7 & 2 & 3 \\ 1 & 2 & 5 & -2 \\ 2 & 3 & -2 & 9 \end{pmatrix}$

Solution

i. $A = \begin{pmatrix} 2 & 3 \\ 3 & 4 \end{pmatrix}$, $A^T = \begin{pmatrix} 2 & 3 \\ 3 & 4 \end{pmatrix}$

ii. $B = \begin{pmatrix} 3 & 1 & 4 \\ 1 & 5 & 2 \\ 4 & 2 & -1 \end{pmatrix}$ $B^T = \begin{pmatrix} 3 & 1 & 4 \\ 1 & 5 & 2 \\ 4 & 2 & -1 \end{pmatrix}$

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Transpose and Symmetric Matrices

iii. C = $\begin{pmatrix} 4 & 5 & 1 & 2 \\ 5 & 7 & 2 & 3 \\ 1 & 2 & 5 & -2 \\ 2 & 3 & -2 & 9 \end{pmatrix}$: C^T = $\begin{pmatrix} 4 & 5 & 1 & 2 \\ 5 & 7 & 2 & 3 \\ 1 & 2 & 5 & -2 \\ 2 & 3 & -2 & 9 \end{pmatrix}$

- From the solutions
  - i. A = A^T
  - ii. B = B^T
  - iii. C = C^T
- Hence the matrices A, B and C are **symmetrical**

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Transpose and Symmetric Matrices

Skew-Symmetric Matrix

- It is a square matrices whose transpose is the same as the negation of the original matrix
- Thus, a matrix **A** is said to be symmetric if and only if $\boldsymbol{A^T = -A}$
- Note that, the main diagonal elements of a skew-symmetric matrix are always zero

$$\boldsymbol{a_{ij} = -a_{ji}}$$

- hence $\boldsymbol{a_{ii}}$ **= 0**

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Transpose and Symmetric Matrices

Illustration

i. $A = \begin{pmatrix} 0 & -3 \\ 3 & 0 \end{pmatrix}, \quad A^T = \begin{pmatrix} 0 & 3 \\ -3 & 0 \end{pmatrix} = -A$

ii. $B = \begin{pmatrix} 0 & 1 & -4 \\ -1 & 0 & 2 \\ 4 & -2 & 0 \end{pmatrix}, B^T = \begin{pmatrix} 0 & -1 & 4 \\ 1 & 0 & -2 \\ -4 & 2 & 0 \end{pmatrix} = -B$

iii. $C = \begin{pmatrix} 0 & 5 & -1 & 2 \\ -5 & 0 & 2 & -3 \\ 1 & -2 & 0 & -2 \\ -2 & 3 & 2 & 0 \end{pmatrix}, C^T = \begin{pmatrix} 0 & -5 & 1 & -2 \\ 5 & 0 & -2 & 3 \\ -1 & 2 & 0 & 2 \\ 2 & -3 & -2 & 0 \end{pmatrix} = -C$

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Matrix Operation

Equality of Matrices

- Two matrices are said to be equal if and only if they have the
  - same order (size/dimension)
  - same entries

If $\mathbf{A} = \begin{bmatrix} 2 & 3 & a \\ b & 6 & 1 \end{bmatrix}$, $\mathbf{B} = \begin{bmatrix} 2 & 3 & 9 \\ -3 & 6 & 1 \end{bmatrix}$, and $\mathbf{C} = \begin{bmatrix} 2 & 3 & 9 \\ -3 & 6 & 1 \\ 0 & 0 & 0 \end{bmatrix}$,

then $\mathbf{A} = \mathbf{B}$ if and only if $a = 9$ and $b = -3$, but $\mathbf{A} \neq \mathbf{C}$ and $\mathbf{B} \neq \mathbf{C}$.

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Matrix Operation

- For matrices A $=\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$ and $B = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix}$ to be equal
  - ✓ their corresponding entries should be equal
- thus, $a_{11} = b_{11}$, $a_{12} = b_{12}$,
  $a_{21} = b_{21}$, $a_{22} = b_{22}$

Illustration

A = $\begin{pmatrix} 5 & 2 \\ 8 & 4 \end{pmatrix}$ and $B = \begin{pmatrix} 5 & 2 \\ 8 & 4 \end{pmatrix}$

A = B

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Matrix Operation

Example

Given matrices A=$\begin{pmatrix} 2x-7 & 5 \\ 8 & 12 \end{pmatrix}$ $and\ B = \begin{pmatrix} 9 & y+3 \\ 8 & 12 \end{pmatrix}$

If A = B, find the values of x and y

Solution

$$\begin{pmatrix} 2x-7 & 5 \\ 8 & 12 \end{pmatrix} = \begin{pmatrix} 9 & y+3 \\ 8 & 12 \end{pmatrix}$$

By comparing entries,

2x – 7 = 9: x = 8

y + 3 = 5: y = 2

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Matrix Operation

Example

Given the matrices

$$A = \begin{pmatrix} 2 & 2y+8 & 4 \\ 7 & 3x+y & 7 \\ 8 & 4 & 9 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & -2x+4y & 4 \\ 7 & 6 & 7 \\ 8 & 4 & 9 \end{pmatrix}$$

Find the values x and y if

(i) A = B

(ii) A + B = 0

TRY

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Matrix Operation

Addition and Subtraction

- Two matrices can be added (subtracted) if they have the same order (size/dimension)
- Note that matrices of different orders (sizes) cannot be added or subtracted
- Matrices that can be added/subtracted are said to be ***CONFORMABLE*** for addition (subtraction)
- Matrix addition is commutative (***WHY???***)
- For matrices A and B, $\boldsymbol{A} \pm \boldsymbol{B} = \left[\boldsymbol{a_{ij}} \pm \boldsymbol{b_{ij}}\right]$

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Matrix Operation

- The operations (+/-) are carried on the corresponding entries of the matrices involved

Illustration

Given A = $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ $and\ B = \begin{pmatrix} e & f \\ g & h \end{pmatrix}$

- A + B = $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ + $\begin{pmatrix} e & f \\ g & h \end{pmatrix}$ = $\begin{pmatrix} a+e & b+f \\ c+g & d+h \end{pmatrix}$
- A – B = $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ - $\begin{pmatrix} e & f \\ g & h \end{pmatrix}$ = $\begin{pmatrix} a-e & b-f \\ c-g & d-h \end{pmatrix}$

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Matrix Operation

Example

Given $A=\begin{pmatrix} 1 & 4 \\ -6 & 3 \end{pmatrix}$, $B=\begin{pmatrix} -5 & 7 \\ -3 & 9 \end{pmatrix}$ and $C=\begin{pmatrix} 8 & 0 \\ 11 & 14 \end{pmatrix}$

a. Find

i. A + B ii. B + A iii. A – C iv. C – A

v. A + (B + C) vi. (A + B)+ C

b. What conclusions can you draw for the operations in (i) and (ii), (iii) and (iv) as well as (v) and (vi) above?

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Matrix Operation

Solution

i. A + B= $\begin{pmatrix} 1 & 4 \\ -6 & 3 \end{pmatrix} + \begin{pmatrix} -5 & 7 \\ -3 & 9 \end{pmatrix}$

$= \begin{pmatrix} 1+(-5) & 4+7 \\ -6+(-3) & 3+9 \end{pmatrix}$

$= \begin{pmatrix} -4 & 11 \\ -9 & 12 \end{pmatrix}$

ii. B + A = $\begin{pmatrix} -5 & 7 \\ -3 & 9 \end{pmatrix} + \begin{pmatrix} 1 & 4 \\ -6 & 3 \end{pmatrix} = \begin{pmatrix} -4 & 11 \\ -9 & 12 \end{pmatrix}$

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Matrix Operation

iii. $A - C = \begin{pmatrix} 1 & 4 \\ -6 & 3 \end{pmatrix} - \begin{pmatrix} 8 & 0 \\ 11 & 14 \end{pmatrix}$

$$= \begin{pmatrix} 1-8 & 4-0 \\ -6-11 & 3-14 \end{pmatrix} = \begin{pmatrix} -7 & 4 \\ -17 & -11 \end{pmatrix}$$

iv. $C - A = \begin{pmatrix} 8 & 0 \\ 11 & 14 \end{pmatrix} - \begin{pmatrix} 1 & 4 \\ -6 & 3 \end{pmatrix}$

$$= \begin{pmatrix} 8-1 & 0-4 \\ 11--6 & 14-3 \end{pmatrix} = \begin{pmatrix} 7 & -4 \\ 17 & 11 \end{pmatrix}$$

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Matrix Operation

v. $A + (B + C) = \begin{pmatrix} 1 & 4 \\ -6 & 3 \end{pmatrix} + \left(\begin{pmatrix} -5 & 7 \\ -3 & 9 \end{pmatrix} + \begin{pmatrix} 8 & 0 \\ 11 & 14 \end{pmatrix} \right)$

$$= \begin{pmatrix} 1 & 4 \\ -6 & 3 \end{pmatrix} + \begin{pmatrix} 3 & 7 \\ 8 & 23 \end{pmatrix} = \begin{pmatrix} 4 & 11 \\ 2 & 26 \end{pmatrix}$$

vi. $(A + B) + C = \left(\begin{pmatrix} 1 & 4 \\ -6 & 3 \end{pmatrix} + \begin{pmatrix} -5 & 7 \\ -3 & 9 \end{pmatrix} \right) + \begin{pmatrix} 7 & -4 \\ 17 & 11 \end{pmatrix}$

$$= \begin{pmatrix} -4 & 11 \\ -9 & 12 \end{pmatrix} + \begin{pmatrix} 8 & 0 \\ 11 & 14 \end{pmatrix} = \begin{pmatrix} 4 & 11 \\ 2 & 26 \end{pmatrix}$$

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Matrix Operation

b.

- A + B = B + A
  - ✓ addition of matrices is commutative

- A – C $\neq C - A$,
  - ✓ subtraction of matrices is not commutative

- A + (B + C) $= (A + B) + C$
  - ✓ addtion is associative

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Matrix Operation

Example

Given $A=\begin{pmatrix} 2 & 8 & 4 \\ 7 & 3 & 7 \\ 8 & 4 & 9 \end{pmatrix}$, $B=\begin{pmatrix} 5 & -7 & 9 \\ 0 & 6 & -7 \\ 3 & 1 & 3 \end{pmatrix}$ and $C = \begin{pmatrix} -6 & 10 \\ 8 & 12 \\ 6 & 4 \end{pmatrix}$

Find

(i) A + B ii. B – A, iii. A - C

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Matrix Operation

Solution

i. A + B = $\begin{pmatrix} 2 & 8 & 4 \\ 7 & 3 & 7 \\ 8 & 4 & 9 \end{pmatrix} + \begin{pmatrix} 5 & -7 & 9 \\ 0 & 6 & -7 \\ 3 & 1 & 3 \end{pmatrix}$

$= \begin{pmatrix} 7 & 1 & 13 \\ 7 & 9 & 0 \\ 11 & 5 & 12 \end{pmatrix}$

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Matrix Operation

ii. B – A = $\begin{pmatrix} 5 & -7 & 9 \\ 0 & 6 & -7 \\ 3 & 1 & 3 \end{pmatrix} - \begin{pmatrix} 2 & 8 & 4 \\ 7 & 3 & 7 \\ 8 & 4 & 9 \end{pmatrix}$

$$= \begin{pmatrix} 3 & -15 & 5 \\ -7 & 3 & -14 \\ -5 & -3 & -6 \end{pmatrix}$$

iii. A – C = $\begin{pmatrix} 2 & 8 & 4 \\ 7 & 3 & 7 \\ 8 & 4 & 9 \end{pmatrix} - \begin{pmatrix} -6 & 10 \\ 8 & 12 \\ 6 & 4 \end{pmatrix}$

- order of A $\neq order\ of\ C$
- the operation is not applicable

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Matrix Operation

TRY

$$A=\begin{bmatrix} 2 & 4 \\ -1 & 3 \end{bmatrix}, B=\begin{bmatrix} 1 & 0 & 3 \\ -2 & 3 & 0 \\ 3 & 0 & 1 \end{bmatrix}, C=\begin{bmatrix} 3 & -3 \\ 2 & 4 \end{bmatrix}$$

Find if possible,

$A+B$ $\qquad$ $A-C$

$A-B$ $\qquad$ $B+A$

$A+C$ $\qquad$ $C+B$

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END OF SLIDES

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Call/Send email to arrange a meeting

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